

Status Epilepticus

Status epilepticus (SE) is a neurological emergency, it is practically defined as 5 min or longer of continuous clinical and/or electrographic seizure activity or recurrent seizure activity without recovery between seizures. The EEG plays a crucial role in the diagnosis, classification, and monitoring treatment of SE [1].

Conceptually, the International League Against Epilepsy (ILAE) suggests that SE results from the failure of mechanisms responsible for seizure termination or initiation of mechanisms which lead to abnormally prolonged seizures (after time point t1) with long-term consequences (after time point t2) that include neuronal injury, death, and alteration of neuronal networks. Hippocampal abnormalities such as atrophy may be evident on neuroimaging [2].

Operationally, the first time point (t1) indicates when emergency treatment of SE should be initiated as it is unlikely to spontaneously terminate beyond this time point and the second time point (t2) indicates when long-term consequences should be expected. The ILAE has published these time points for some types of SE. These general approximations are intended for operational use and based on experimental models; collaborative clinical evidence is emerging (see Table 24.1) [3].

Status epilepticus is described based on semiology (motor features), etiology (acute, remote, or progressive), electrographic features, and the age of occurrence.

Semiologically, it may be classified into two broad types:

1. *SE with prominent motor features*: This includes convulsive SE (CSE) and other less common forms such as focal motor status (epilepsia partialis continua and repetitive focal motor seizures/Jacksonian SE), tonic, clonic, and myoclonic SE.
2. *SE without prominent motor features (nonconvulsive SE)*: This includes focal status with coma or impaired consciousness, absence status (may be typical, atypical, or de novo), aphasic status, and aura continua.

Etiologically, status epilepticus may be symptomatic of acute injury (such as a stroke or encephalitis), remote pathologies (such as poststroke or post-traumatic) or progressive conditions such as tumors or neurodegenerative

Table 24.1 Operational time points for status epilepticus

Type of SE	Time (t1) when a seizure becomes SE	Time (t2) when SE may cause long-term consequences
CSE (tonic clonic)	5 min	Greater than 30 min
NCSE (focal with impaired consciousness)	10 min	Greater than 60 min
NCSE (absence SE)	10–15 min	Unknown

disease. The electrographic features in SE are variable. Not all forms are associated with prominent electrographic correlates. Surface EEG findings are seen in only about 30% of those with pure motor seizures and 20% of those with nonmotor seizures without impaired awareness. Some forms of SE may be an integral part of age-dependent clinical syndromes [4].

Convulsive Status Epilepticus (CSE)

Convulsive status epilepticus may be generalized (GCSE) or more commonly focal onset with secondary generalization (focal to bilateral convulsive SE).

Clinically apparent CSE doesn't require an EEG for diagnosis. In fact, it may be significantly obscured by continuous (tonic phase) or regularly fragmented (clonic phase) myogenic/movement artifact as shown in Chapter 19, Figures 19.2 and 19.3. However, the EEG is crucial to differentiate mimics such as psychogenic status and other nonepileptic conditions with abnormal movements such as tremors, myoclonus, and dyskinesias. The EEG is diagnostic in situations where convulsive activity may not be apparent such as neuromuscular blockade. The EEG enables characterization and localization of ictal activity during CSE. After convulsions cease, it helps identify the cause of persistent altered mentation which may be secondary to sedatives, postictal encephalopathy, or ongoing ictal activity (NCSE) as shown in Figure 24.1. One study of 145 patients with CSE showed that about a third will evolve to NCSE [5]. Additionally, the EEG guides management including adjustments of sedation, instituting burst-suppression and monitoring for refractoriness or recurrence.

Initially, discrete electrographic seizures with interval slowing may occur, later these merges with waxing and waning ictal discharges or rhythmic ictal activity that may become continuous. Figure 24.2 shows focal status epilepticus while Figure 24.3 shows generalized status epilepticus.

As SE progresses, brief periods of suppression interrupt the continuous ictal activity as shown in Figure 24.4. Termination of ictal activity is typically

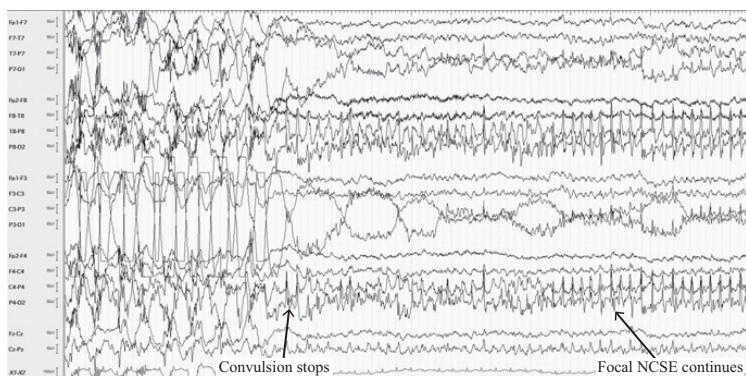


Figure 24.1 Twenty-seven-year-old woman with persistent postictal confusion, persistent right posterior temporal ictal activity consistent with NCSE.

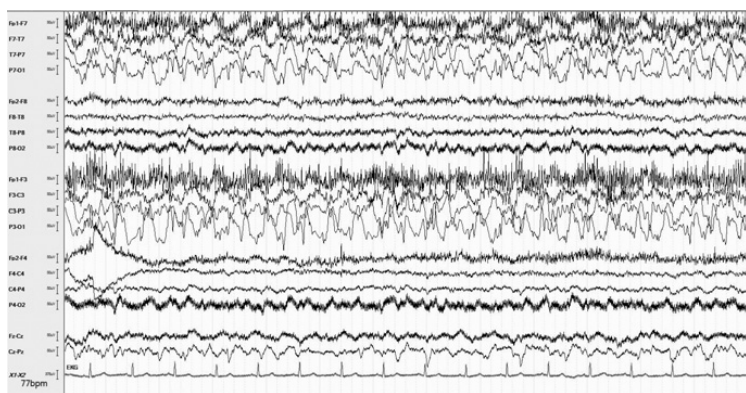


Figure 24.2 Forty-eight-year-old woman with CNS lymphoma, focal status epilepticus of left hemispheric onset.

characterized by generalized or lateralized periodic discharges, background suppression, and/or slowing with a gradual restoration of normal activities. Variations to these patterns are common and these stages may not follow in sequence. Cardiac rhythm disturbances such as ictal or postictal asystole may be noted on single EKG channel as previously discussed in Chapter 18 [6].

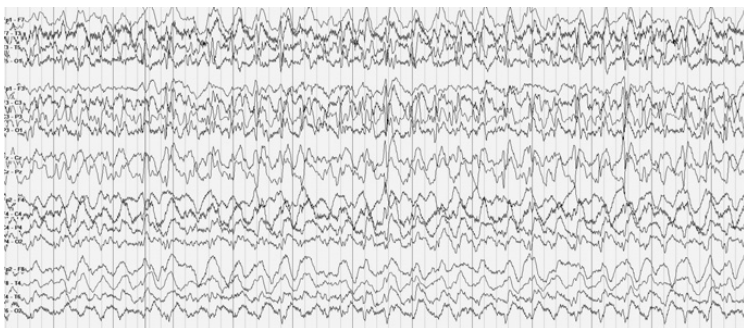


Figure 24.3 Same patient as in Figure 24.2, secondary generalized status epilepticus.

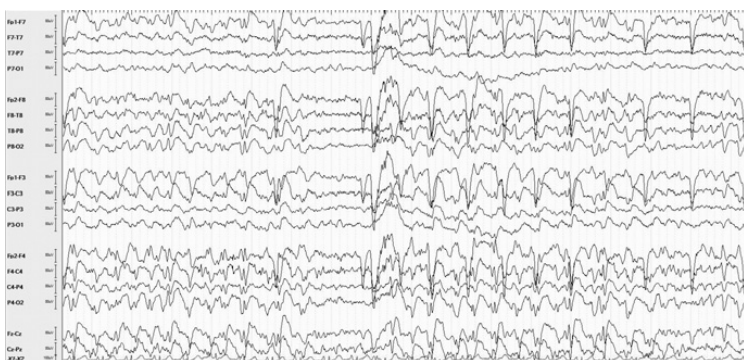


Figure 24.4 Fifty-six-year-old man with encephalitis, brief suppressions interrupt continuous ictal activity.

Nonconvulsive Status Epilepticus (NCSE)

Nonconvulsive SE is characterized by continuous or repetitive electrographic seizures without prominent motor features. Mentation may be affected to a variable degree ranging from subtle impairments in awareness or behavior to unresponsiveness and coma. This may occur following focal onset seizures that spread to impair awareness in ambulatory adults such as those with temporal or frontal lobe epilepsies, following CSE after convulsions have ceased or occur spontaneously in hospitalized and critically ill individuals with systemic and/or cerebral dysfunction

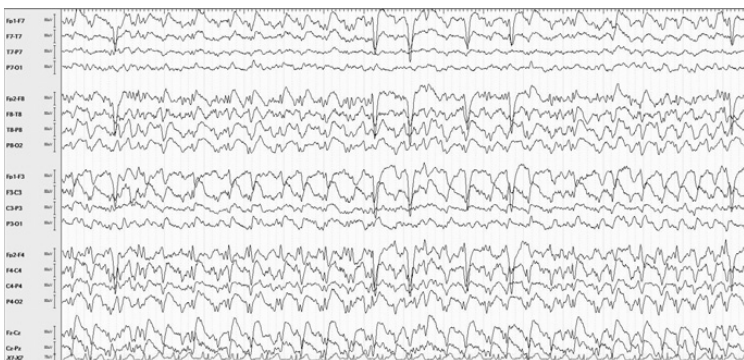


Figure 24.5 Same patient as in Figure 24.4, right hemispheric rhythmic polyspikes and wave discharges with wide fields consistent with focal onset NCSE.

who may not have a prior history of epilepsy. Since accompanying clinical manifestations may be subtle or absent, the EEG is crucial for diagnosis. Multiple studies have shown the incidence of NCSE to be at least 10–20% of those undergoing continuous EEG monitoring [7]. The diagnosis and clinical implications of electrographic seizures have been discussed in Chapter 16 (ictal patterns). Figure 24.5 shows NCSE.

Other Less Common Forms of Status Epilepticus

Epilepsia Partialis Continua (EPC)

Epilepsia partialis continua consists of focal onset motor SE (without impaired awareness) characterized by segmental myoclonus that may persist for prolonged periods. Common causes include strokes, malignancies, encephalitis, nonketotic hyperglycemia, or neuronal migration defects. The EEG is typically normal, electrographic changes occur in less than a third of patients and typically consist of focal discharges that correlate with contralateral motor movements as shown in Figure 24.6. This type of SE is often refractory to pharmacological treatments [8].

Absence Status

Absence SE may occur in three distinct forms:

1. *Typical absence SE*: This is seen in patients with idiopathic generalized epilepsies (IGE) such as CAE, JAE, or JME and is characterized by mild

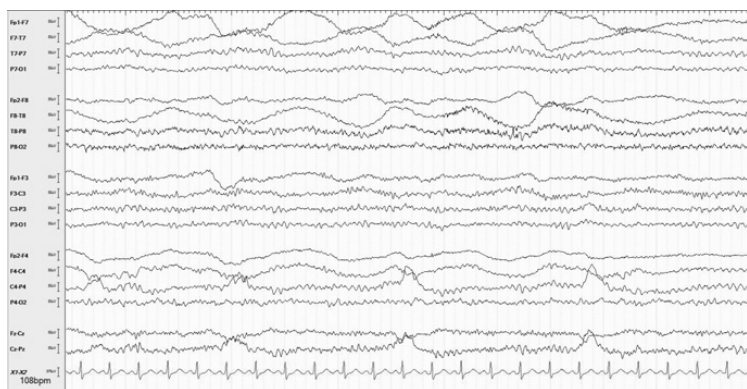


Figure 24.6 Fifty-nine-year-old woman with a right cortical stroke, focal right paracentral (C4) blunted LPDs associated with repetitive left arm twitching.

impairments of awareness or confusional states, blinking, and occasional myoclonus that may persist for several hours. Sleep deprivation, alcohol, and benzodiazepine withdrawal are common triggers. A heralding convulsion may occur at the onset or terminate the event. EEG shows continuous generalized spike (polyspikes) and wave discharges up to 4 Hz in frequency with a frontal or central predominance. Spontaneous termination is not uncommon.

2. *Atypical absence SE*: This is relatively rare, typically occurring in infants and adolescents with underlying developmental deficits. Clinically it is characterized by confusion, subtle myoclonus, tonic, and atonic seizures. The EEG shows slow spike wave pattern and diagnosis typically requires careful comparison with prior studies.
3. *De novo absence SE*: This is characterized by confusion and is seen in older adults in the setting of abrupt withdrawals of benzodiazepines or other sleeping aids. Prior IGE may be present. EEG is like typical absence SE. Easily responds to treatment with benzodiazepines [9].

Tonic Status Epilepticus

Generalized Tonic SE is usually associated with Lennox–Gastaut syndrome (LGS). It is characterized by repetitive brief tonic contractions of the arms and extension of the legs. The EEG usually shows diffuse low-voltage fast activity, polyspikes, and slow waves as shown in Figure 24.7.

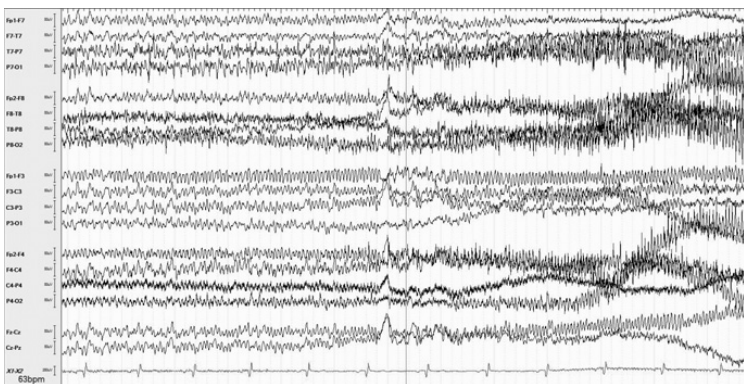


Figure 24.7 Thirty-one-year-old woman with LGS, continuous generalized low-amplitude fast activity associated with tonic status epilepticus.

Clonic Status Epilepticus

Generalized Clonic SE is usually seen in infants and children in association with febrile seizures or LGS. The EEG shows bilateral synchronous spikes, spiky bursts, and/or spike wave complexes.

Atonic Status Epilepticus

Atonic SE is usually seen in infants and children. It is clinically characterized by minor jerking, eye movements, and unresponsiveness. The EEG shows bilateral synchronous spike and slow-wave complexes. Prognosis is generally favorable.

Myoclonic Status Epilepticus

Myoclonus may be cortical (epileptic) or subcortical in origin (nonepileptic). In cortical myoclonus, myoclonic jerks are associated with epileptic discharges. Myoclonic status epilepticus refers to continuous epileptic myoclonus and potentially other seizure types. This is rarely seen in children with idiopathic generalized epilepsies (IGE) where the prognosis is usually favorable. This should be differentiated from myoclonic status epilepticus that occurs post cardiac arrest (sometimes referred to as status myoclonicus) which is typically associated with a poor prognosis. The EEG shows bilateral synchronous discharges associated with myoclonic artifact; these are best appreciated in the central channels that are relatively free of muscle artifact. Neuromuscular blockade may be selectively used to better appreciate epileptic

discharges in those where the record is obscured with excessive myogenic artifact. Subcortical myoclonus such as metabolic myoclonus or that seen in posthypoxic syndrome (Lance Adam's syndrome) may not have reliable electrographic correlates [10]. Common electrographic patterns in cerebral anoxia are further described in Chapter 25.

Chapter Summary

1. SE is defined as 5 min or longer of continuous clinical and/or electrographic seizure activity or recurrent seizures without interval recovery.
2. A t1 refers to the time point beyond which there is failure of mechanisms responsible for seizure termination or initiation of mechanisms, which lead to abnormally prolonged seizures.
3. A t2 refers to the time point beyond which there are long-term consequences due to neuronal injury, death, and alteration of neuronal networks.
4. Semiologically, SE can be classified as with or without prominent motor features.
5. Convulsive SE may evolve into NCSE in a significant minority after convulsive activity ceases.
6. NCSE may be diagnosed on EEG in a significant minority of critically ill patients.
7. EPC may not be associated with ictal activity on surface EEG.
8. De novo absence SE may be seen in older individuals in the setting of abrupt benzodiazepine withdrawal. They may have a previous or family history of IGE.

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